Co-evolution of face-recognition and naming: Unifying various functions of the fusiform gyrus brain area

Ljiljana Progovac*

*Corresponding Author: progovac@wayne.edu
Department: Linguistics, Wayne State University, Detroit, USA

Human face recognition is not only expert-like (quick and precise), but it is also highly localized and lateralized, specifically in the right fusiform gyrus area. This was proposed to be a result of recent human evolution, due to the expansion of regions engaged in language. This same fusiform gyrus area is also implicated in naming, in visual processing, in metaphorical language, and, also in processing simple syntactic chunks. I provide evidence for the proposal that enhanced face recognition in humans was facilitated by (co-evolved with) the early emergence of language, specifically the early naming strategies using simple syntactic chunks to name people in memorable ways, consistent with the memory abilities of our ancestors at the time. In this respect, the fusiform face area is activated more strongly by fearful faces than neutral faces, with emotion playing a role. An fMRI experiment found that the processing of such emotive, typically derogatory names/nicknames strongly implicates the right fusiform gyrus area. Moreover, recent experiments found a strong advantage for face individuation for (named) familiar over unfamiliar individuals, as well as a strong correlation between better face-name recall and better face recognition ability.

1. Specialized brain networks for expert face recognition in humans

Humans have evolved highly specialized brain networks for recognizing faces, where the fusiform gyrus area, specifically the right fusiform gyrus, plays a crucial role. Typical human adults are considered to be experts at individual face recognition (Carey, 1992; McKone, et al., 2007; Tanaka, 2001; Young & Burton, 2018), but see section 4 for additional atypical cases. According to Rossion & Taubert (2019), human face recognition of conspecifics is not only expert-like (quick and precise), but it is also highly localized and lateralized, leading to selective disorders such as prosopagnosia (face blindness) in case of focal damage to these brain regions.[1] In this respect, primates, including well-studied rhesus

[1] Prosopagnosia is a severe and specific breakdown of the ability to recognize individual faces (Bodamer, 1947; Davies-Thompson, et al., 2014; Hecaen & Angelergues, 1962; Meadows, 1974). These ventral face-selective regions respond to faces significantly more strongly compared to other stimuli (Kanwisher, et al., 1997; Fu, et al., 2012; Grill-Spector, et al., 2018; Palsamudram, et al., 2024).

monkeys, seem to show much reduced abilities for individual face recognition of conspecifics (Rossion and Taubert, 2019), suggestive of a qualitative difference (e.g. McKone, et al., 2007; Rossion, 2008).[2] It is clear why such expert face recognition would be highly beneficial, in fact adaptive, for human beings, who live in societies with complex interpersonal dynamics. The question is what facilitated the enhancement of this face recognition mechanism in humans to such an expert degree.

2. What's in a name? Co-evolution of naming and face recognition

As pointed out above, the fusiform gyrus, in particular right BA 37, is directly implicated in face perception and recognition (Kanwisher, et al., 1997; Grill-Spector, et al., 2017; Rapcsak, 2019; Barton, 2022). This same area is also implicated in visual processing, concreteness, metaphorical language, and, importantly, naming (see e.g. a meta-analysis by Ardila, et al., 2015). The fusiform gyrus is also of relevance for processing simple syntactic chunks (section 3). Here I argue that the involvement of the fusiform gyrus area in humans in both face recognition and in naming is not to be seen as a coincidence. Rather, the enhanced face recognition in humans was facilitated by the early emergence of language (see Benítez-Burraco & Progovac, 2024), specifically by the early simple syntactic strategies for naming people in memorable ways, consistent with the memory abilities of our ancestors at that time. There is evidence for the prediction of this proposal that there should exist interactions and correlations between naming and face recognition, both in neurotypical and atypical populations (see also section 4).

In this respect, it is telling that one of the characteristics of face individuation in humans is the strong advantage found for familiar over unfamiliar individuals (e.g., Bruce, 1982; Burton, et al., 1999; Bruce, et al., 2001; Megreya & Burton, 2006; Young & Burton, 2018). Another study found that there is stronger activity in the fusiform face area when a person sees a familiar face as opposed to an unfamiliar one (Weibert & Andrews, 2015).[3] Using the fMRI method, these authors also found that the participants that were more accurate in identifying familiar faces had more activity in their right fusiform face area. Some authors have claimed, on the basis of the robust advantage found for familiar

[2] As pointed out by Rossion & Taubert (2019), other species often use other clues, such as smell or vocalizations, or a combination of clues, to recognize conspecifics.

[3] The 'fusiform face area' (FFA), as named by Kanwisher et al. (1997), is located in the fusiform gyrus (BA 37) area, and is considered to be the region of the most consistent and robust face selective activation (see also Kanwisher and Yoval 2006).

faces, that human expertise at individual face recognition may concern only familiar faces (Young & Burton, 2018; but see Rossion & Taubert, 2019).[4] Given that familiar faces are much more likely to be associated with (familiar) names, this finding is consistent with the proposal that using language to name/label people contributes to the ease of recognition/individuation. Moreover, language processing in spoken conversation is not limited only to the sound, but it also significantly depends on facial and other visual cues, including the positioning of the lips (consider e.g. the McGurk effect). When our ancestors started using spoken language in conversation, it is conceivable that they paid much more attention to the faces of their conspecifics than they did prior to that. [5]

Clear support for this proposal comes from a recent experiment with 99 neurotypical individuals, which found a correlation between better face-name recall and better face recognition ability, concluding that "the ability to learn and recall proper names is particularly important to face recognition" (DeGutis, et al., 2024). If naming is helpful, the question is what kind of naming would have worked best in early human evolution. Interestingly, some recent findings indicate that the fusiform face area has other functions which are implicated in emotion: it is differentially activated by faces exhibiting different emotions. As one example, the fusiform face area is activated more strongly by fearful faces than neutral faces (Guyer, et al. 2010). This can be taken to mean that naming which stirs emotions would be more noticed and better etched in memory than naming which is purely labelling. This is significant because the memory capacities of our ancestors at the point when language just started to emerge may not have been potent enough to retain a series of random labels that would have constituted people's names. But if there were naming strategies that appealed to emotion, which were also descriptive of people and their actions, they would have had a better chance of being memorized. The following section provides a specific proposal regarding such early strategies for naming, supported by a neuroimaging experiment.

3. An fMRI study of language evolution implicates the fusiform gyrus area

An fMRI study found that the processing of ancestral-like (flat) verb-noun compounds (illustrated below), which are typically used for (derogatory) naming and nicknaming across different languages and cultures, shows enhanced

[4] In this respect, a rare study that compared familiar and unfamiliar face matching in macaques failed to find an advantage for familiar faces (Micheletta, et al., 2015).

[5] Interestingly, a bouba–kiki effect has been identified as well, in which people associate proper names featuring rounded sounds (e.g. Bob) with round-faced (versus angular-faced) individuals (see e.g. Barton & Halberstadt, 2018), highlighting another link between language and face perception.

activation in the right fusiform gyrus area (BA 37), the area that is directly implicated in both naming and face recognition (Progovac, et al., 2018). As pointed out in e.g., Weibert & Andrews (2015), face recognition is facilitated primarily by the right (as opposed to left) BA 37, which was also the finding in this experiment. These flat(ter) compounds, in contrast to their more hierarchical counterparts, resulted in increased activation in the inferior temporal gyrus and the right fusiform gyrus (Brodmann's areas 37/19). These verb-noun compositions have been proposed to be proxies or "living fossils" of the earliest stages of language evolution (see Jackendoff, 1999; Progovac, 2015, 2016).

Moreover, the examples of this naming strategy preserved across languages tend to be highly expressive and metaphorical, as well as derogatory when referring to humans, which is certainly relevant for the emotional dimension of these creations, and for their memorability.[6]

(1) English: kill-joy, turn-skin, turn-coat, hunch-back, wag-tail, tattle-tale, scatter-brain, cut-throat, mar-wood (bad carpenter), heck-wood, busy-body, cry-baby, break-back, fill-belly (glutton), lick-spit, pinch-back (miser), spit-fire, tumble-dung (insect), crake-bone (crack-bone), shave-tail (shove-tail), wipe-tail, fuck-ass, fuck-head, shit-ass, shit-head

(2) Serbian: cepi-dlaka 'split-hair' (hair-splitter); guli-koža 'peel-skin' (who rips you off); vrti-guz 'spin-butt' (restless person, fidget); muti-voda 'muddy-water' (trouble-maker); jebi-vetar 'fuck-wind' (charlatan); vuci-guz 'drag-butt' (slow-moving person); gori-guzica 'burn-butt' (a person in trouble, burn-breeches); kosi-noga 'skew-leg' (person who limps); jedi-vek, 'eat-life' (one who constantly annoys); podvi-rep 'fold-tail' (one who is crestfallen); deri-muda 'rip-balls' (place name, a steep hill); gladi-kur 'stroke-dick' (womanizer); kapi-kur 'drip-dick' (name of a slow water spring); liz-guz 'lick-butt;' piš-kur 'piss-dick;' plači-guz 'cry-butt;' poj-kurić 'sing-dick' (womanizer); seri-vuk 'shit-wolf'

The proxies of these ancestral verb-noun compositions represent a rather fertile strategy for naming/nicknaming. According to Weekley's (1916) book titled *Surnames*, devoted specifically to these compounds in English, there were thousands of such creations in medieval times, constituting an "expressive way of *naming*," but many fail to be preserved in dictionaries or grammar books due to their "unquotable coarseness," with frequent references to body parts and

[6] These examples were cited in Progovac & Locke (2009) and Progovac (2015, 2016), with most of them ultimately taken from Mihajlović (1992) and Weekley (1916). Hundreds more can be found in these references, including a detailed syntactic analysis in Progovac (2015).

functions. For Darmesteter (1934, 443), the artistic beauty and richness of these compounds [in French] is inexhaustible: "it would be well could French poets again make use in lofty poetry of this class of *epithets*, for they may attain Homeric breadth…" Mihajlović, who devoted his career to collecting over 500 Serbian people and place *names* in the form of these compounds, calls them condensed compositions which pack in them … frozen fairy tales, proverbs, and ancient wisdoms and metaphors (1992, 8–9).

Such compound creations would have certainly been memorable, standing a better chance of being used and reused to refer to a person, essentially starting a naming tradition. In recent history, when last names just started to be used, many such nicknames (often with a change in a sound or two, to mask their derogatory nature) became last names that run in families (see e.g. Mihajlović, 1992; Weekley, 1916). Extrapolating from this practice, it is plausible that a similar process was in place for one's first (and only) names, at the point when language/grammar just started to emerge, and people did not receive their first names at birth. This simple proto-syntactic strategy for naming would have been exactly the right kind of phenomenon to engage the fusiform gyrus area and to spiral the co-evolution of the cultural process of naming and the genetic predisposition for expert face recognition and individuation in humans.[7] Recall that the fusiform gyrus area is directly implicated in both naming and face recognition, but also in concreteness, metaphors and emotion. The hypothesized ancestral proto-syntactic compositions of just one verb-like and one noun-like element engage all these aspects, shedding light on why the fusiform gyrus area would be implicated in all these phenomena, which otherwise appear to be unrelated.

There is in fact one more connection between the fusiform gyrus area and the verb-noun composition strategy, the strategy which is, by the way, at the core of all human grammars. The fusiform gyrus region also seems to be involved in noun-verb classification (Boylan, et al., 2014; Benítez-Burraco & Progovac, 2024). According to Boylan, et al. (2014), differentiating between nouns and verbs in the fusiform gyrus areas may arise from syntactic rather than semantic cues, because this area seems unable to classify nouns vs. verbs when they are presented isolated. Related to that, the fusiform gyrus region has been associated with the generation of simple (syntactic) memory chunks (Bonhage, et al., 2017),

[7] Advocating co-evolution of language and emotions, Jablonka, et al. (2012: 2152) also ascribed an active, causal role to language in human evolution, stating that the emergence of some genuinely new cultural/linguistic practices "drove a process of genetic accommodation of both general and language-specific aspects of cognition."

and it is considered to be at the bottom of a hierarchy of computational resources recruited for processing increasingly complex syntactic structures (e.g. Makuuchi & Friederici, 2013). This constitutes one more reason for the robust activation of the fusiform gyrus during the processing of the ancestral, syntactically simplest verb-noun compositions.

An evolutionary scenario supportive of the co-evolution of language and face recognition is mentioned in Rossion & Taubert (2019). For them, the anatomico-functional differences between species are consistent with the possibility that the ventral face-selective recognition system – which is found essentially all along the fusiform gyrus and adjacent sulci – is the result of recent human evolution, involving the expansion of regions engaged in language and social cognition.

4. Testing potential: Cognitive conditions implicating the fusiform gyrus

This proposal opens the possibility to test the involvement of the fusiform gyrus region in naming/nicknaming and face recognition, specifically trying to determine to what extent these two processes interact/overlap in neurotypical and atypical populations, including prosopagnosia, but also autism (ASD) and synesthesia. Studies found that ASD shows lower neuron densities in the fusiform face area (van Kooten, et al., 2008), and this may be one of the contributing factors both to the impairments in face processing (e.g. Campatelli, et al., 2013; Weigelt, et al., 2022; Golarai, et al., 2006), and to the altered processing of names (see e.g., Nowicka, et al.'s 2016 review). The fusiform gyrus area is also implicated in synesthesia (e.g., van Leeuwen, et al., 2011), and there are also reports, in need of further scrutiny, that synesthetes may outperform controls in fine visual discrimination of facial identity and emotion (Janik-McErlean, et al., 2016). According to Ramachandran & Hubbard (2001), the beginning of cross-modality (crucial for language evolution) involved a cross-wiring in the fusiform gyrus. Given the proposal in this paper, what contributed to this cross-wiring would have been the emergence of early forms of language for naming, showing metaphoricity, imageability, and emotion, and contributing to the robust enhancement of face recognition and individuation in humans.

Acknowledgements

I am grateful for the constructive comments provided by the reviewers.

References

Ardila, A., Bernal, B., & Rosselli, M. (2015). Language and visual perception associations: Meta-analytic connectivity modeling of Brodmann area 37. *Behavioural Neurology,* 1–14. doi: 10.1155/2015/565871

Barton, J. J. S. (2022). Face processing in the temporal lobe. *Handbook of Clinical Neurology,* 187, 191–210. doi: 10.1016/B978-0-12-823493-8.00019-5

Barton, D.N., & Halberstadt, J. (2018). A social Bouba/Kiki effect: A bias for people whose names match their faces. *Psychonomic Bulletin and Review*, 25(3), 1013-1020. doi: 10.3758/s13423-017-1304-x. PMID: 28597194.

Benítez-Burraco, A., & Progovac, L. (2024). Syntax and the brain: Language evolution as the missing link(ing theory)? *Frontiers in Psychology,* 15, 1445192. doi: 10.3389/fpsyg.2024.1445192

Bodamer, J. (1947). Die Prosopagnosie. *Archiv fur Psychiatrie und Zeitschrift fur Neurologie*, 179, 6–54.

Bonhage, C. E., Meyer, L., Gruber, T., Friederici, A. D., & Mueller, J. L. (2017). Oscillatory EEG dynamics underlying automatic chunking during sentence processing. *NeuroImage* 152, 647–657. doi: 10.1016/j.neuroimage.2017.03.018

Boylan, C., Trueswell, J. C., & Thompson-Schill, S. L. (2014). Multi-voxel pattern analysis of noun and verb differences in ventral temporal cortex. *Brain and Language,* 137, 40–49. doi: 10.1016/j.bandl.2014.07.009

Bruce, C. (1982). Face recognition by monkeys: Absence of an inversion effect. *Neuropsychologia*, 20, 515–521.

Bruce, V., Henderson, Z., Newman, C., & Burton, A. M. (2001). Matching identities of familiar and unfamiliar faces caught on CCTV images. *Journal of Experimental Psychology: Applied*, 7, 207–218.

Burton, A. M., Wilson, S., Cowan, W., & Bruce, V. (1999). Face recognition in poor-quality video: Evidence from security surveillance. *Psychological Science*, 10, 243–248.

Campatelli G., Federico R.R., Apicella F., Sicca F., & Muratori, F. (2013). Face processing in children with ASD: Literature review. *Research in Autism Spectrum Dis*order, 7, 444–54.

Carey, S. (1992). Becoming a face expert. *Philosophical Transactions of the Royal Society of London B*, 335, 95–103.

Darmesteter, A. (1934). *A Historical French grammar.* Authorized English edition by Alphonse Hartog. First edition in 1899. London: Macmillan and Co.

Davies-Thompson, J., Pancaroglu, R., & Barton, J. (2014). Acquired prosopagnosia: Structural basis and processing impairments. *Frontiers in Bioscience* (Elite Ed.), 6, 159–174.

DeGutis, J., Palsamudram, T., Campbell, A., Fry, R., Verfaellie, M., & Anderson, N. D. (2024). Better face-name recall is associated with better face recognition ability. *Quarterly Journal of Experimental Psychology*, *78*(9), 1769-1784. https://doi.org/10.1177/17470218241290229

Fu, S., Feng, C., Guo, S., Luo, Y., & Parasuraman, R. (2012). Neural adaptation provides evidence for categorical differences in processing of faces and Chinese characters: An ERP study of the N170. *PLoS ONE* 7(7), e41103. doi:10.1371/journal.pone.0041103

Golarai, G., Grill-Specto, K., & Reiss, A.L. (2006). Autism and the development of face processing. *Clinical Neuroscience Res*earch, 6, 145–60.

Grill-Spector, K, Weiner, K.S., Kay, K., & Gomez, J. (2017). The functional neuroanatomy of human face perception. *Annual Review of Vision Science*, 3, 167–196. doi:10.1146/annurev-vision-102016-061214

Grill-Spector, K., Weiner, K.S., Gomez, J., Stigliani, A., & Natu, V.S. (2018). The functional neuroanatomy of face perception: From brain measurements to deep neural networks. *Interface Focus* 8, 20180013. http://dx.doi.org/10.1098/rsfs.2018.0013

Guyer, A.E., Monk, C.S., McClure-Tone, E.B., Nelson, E.E., Roberson-Nay, R., Adler, A.D., Fromm, S.J., Leibenluft, E., Pine, D.S., & Ernst, M. (2010). A developmental examination of amygdala response to facial expressions. *Journal of Cognitive Neuroscience*, 20 (9), 156582. doi:10.1162/jocn.2008.20114. PMC 2902865 PMID 18345988

Hecaen, H., & Angelergues, R. (1962). Agnosia for faces (prosopagnosia). *Archives of Neurology*, 7, 92–100.

Jablonka, E., Ginsburg, S., & Dor, D. (2012). The co-evolution of language and emotions. *Philosophical Transactions of Royal Society B: Biological Sciences,* 367, 2152–2159. doi: 10.1098/rstb.2012.0117

Jackendoff, R. (1999). Possible stages in the evolution of the language capacity. *Trends in Cognitive Sciences,* 3, 272–279. doi: 10.1016/S1364-6613(99)01333-9

Janik McErlean, A.B., Susilo, T., Rezlescu, C., Bray, A., & Banissy, M.J. (2016). Social perception in synaesthesia for colour. *Cognitive Neuropsychol*ogy, 33(7-8), 378-387. doi: 10.1080/02643294.2016.1261820. Epub 2016 Dec 11. PMID: 27937073.

Kanwisher, N., McDermott, J., & Chun, M.M. (1997). The fusiform face area: A module in human extrastriate cortex specialized for face perception. *Journal of Neuroscience,* 17 (11): *4302–11. doi:10.1523/JNEUROSCI.17-11-04302.1997. PMC 6573547. PMID 9151747*

Kanwisher N, &mYovel G. (2006). The fusiform face area: a cortical region specialized for the perception of faces. *Philosophical Transactions of Royal Society B*. 29;361(1476):2109-28. doi: 10.1098/rstb.2006.1934. PMID: 17118927; PMCID: PMC1857737.

Makuuchi, M., & Friederici, A. D. (2013). Hierarchical functional connectivity between the core language system and the working memory system. *Cortex; A Journal Devoted to the Study of the Nervous System and Behavior,* 49, 2416–2423. doi: 10.1016/j.cortex.2013.01.007

McKone, E., Kanwisher, N., & Duchaine, B. C. (2007). Can generic expertise explain spezial processing for faces? *Trends in Cognitive Sciences*, 11, 8–15.

Meadows, J. C. (1974). The anatomical basis of prosopagnosia. *Journal of Neurology, Neurosurgery, and Psychiatry*, 37, 489–501.

Megreya, A. M., & Burton, A. M. (2006). Unfamiliar faces are not faces: Evidence from a matching task. *Memory & Cognition*, 34, 865–876.

Micheletta, J., Whitehouse, J., Parr, L. A., Marshman, P., Engelhardt, A., & Waller, B. M. (2015). Familiar and unfamiliar face recognition in crested macaques (Macaca nigra). *Royal Society Open Science*, 2(5), 150109.

Mihajlović, V. (1992). *Ime po zapovesti* (*Name by command*). Beograd: Nolit.

Nowicka, A., Cygan, H.B., Tacikowski, P., Ostaszewski, P., & Kuś, R. (2016). Name recognition in autism: EEG evidence of altered patterns of brain activity and connectivity. *Molecular Autism* 7, 38. https://doi.org/10.1186/s13229-016-0102-z

Palsamudram, T. Campbell, A., Fry, R., Yosef, B., Kirsch, L., Anderson, N.D., Verfaellie, M., & DeGutis, J. (2024). Face naming and recollection represent key memory deficits in developmental prosopagnosia. *Cortex,* 180, 78-93. doi.org/10.1016/j.cortex.2024.08.003.

Progovac, L. (2015). *Evolutionary Syntax*. Oxford Studies in the Evolution of Language. Oxford: Oxford University Press. doi.org/10.1093/acprof:oso/9780198736547.001.0001

Progovac, L. (2016). A Gradualist scenario for language evolution: Precise linguistic reconstruction of early human (and Neandertal) grammars. *Frontiers in Psychology*, 7, 1714. doi.org/10.3389/fpsyg.2016.01714

Progovac, L., & Locke, J. L. (2009). The urge to merge: Ritual insult and the evolution of syntax. *Biolinguistics,* 3.2–3, 337–54. doi.org/10.5964/bioling.8707

Progovac, L., Rakhlin, N., Angell, W., Liddane, R., Tang, L., & Ofen, N. (2018). Neural correlates of syntax and proto-syntax: An fMRI study. *Frontiers in Psychology,* 9, 1–16. doi: 10.3389/fpsyg.2018.02415

Ramachandran, V.S., & Hubbard, E.M. (2001). Synaesthesia: A window into perception, thought and language. *Journal of Consciousness Studies*, 8, 3–34.

Rapcsak, S. Z. (2019). Face recognition. *Current Neurology and Neuroscience Reports,* 19, 41. doi: 10.1007/s11910-019-0960-9

Rossion, B. (2008). Picture-plane inversion leads to qualitative changes of face perception. *Acta Psychologica*, 128, 274–289.

Rossion, B. & Taubert, J. (2019). What can we learn about human individual face recognition from experimental studies in monkeys? *Vision Research,* 157, 142-158. doi.org/10.1016/j.visres.2018.03.012

Tanaka, J. W. (2001). The entry point of face recognition: Evidence for face expertise. *Journal of Experimental Psychology: General*, 130, 534–543.

van Kooten, I.A., Palmen, S.J., von Cappeln, P., Steinbusch, H.W., Korr, H., Heinsen, H., Hof, P.R., van Engeland, H., & Schmitz, C. (2008). Neurons in the fusiform gyrus are fewer and smaller in autism. *Brain,* 131 (4), 987-99. doi:10.1093/brain/awn033. PMID 18332073

van Leeuwen, T. M., den Ouden, H.E.M., & Hagoort, P. (2011). Effective connectivity determines the nature of subjective experience in grapheme-color synesthesia. *The Journal of Neuroscience,* 31(27), 9879-84. doi: 10.1523/JNEUROSCI.0569-11.2011

Vrticka, P., Black, J.M., & Reiss, A.L. (2013). The neural basis of humour processing. *Nature Reviews Neuroscience*, 14, 860–868. doi:10.1038/nrn3566

Weekley, E. (1916). *Surnames*. New York: E.P. Dutton and Co.

Weibert, K., & Andrews, T.J. (2015). Activity in the right fusiform face area predicts the behavioural advantage for the perception of familiar faces. *Neuropsychologia*, 75, 58896. doi:10.1016/j.neuropsychologia.2015.07.015. PMID 26187507. S2CID 17278436.

Weigelt, S., Koldewyn, K., & Kanwisher, N. (2012). Face identity recognition in autism spectrum disorders: A review of behavioral studies. *Neuroscience and Biobehavioral Reviews*, 36, 1060–84.

Young, A. W., & Burton, A. M. (2018). Are we face experts? *Trends in Cognitive Sciences*, 22, 100–110.